ABHIJAY GANGARAPU

Incoming Freshman | McNeil High School

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EDUCATION

McNeil High School ***Austin, TX***

Class of 2030 | Expected Graduation: May 2030

- Incoming Freshman (completed 8th grade)
- Two grades ahead in mathematics

RESEARCH INTERESTS

Number Theory • Algebra • p-adic Analysis • Algebraic Structures • Applied Mathematics • Computational Mathematics • Cryptography • Cybersecurity

PROJECTS

Algebraic Surjectivity and Super-Linear Period Growth in p-adic Piecewise Affine Maps

Independent Research | Number Theory & Neural Learning Limits ***2025–2026***

- Proved monodromy-based lift formula for period growth in piecewise affine maps over $\mathbb{Z}/p^k\mathbb{Z}$ with Hensel lifting
- Identified fundamental learning threshold $T/L_{in} \approx 35$–150 where gradient-based neural networks fail on long-period algebraic sequences
- Validated across 168 primes ($p = 2$ to $p = 997$) with mean growth ratio 26.94× (maximum 79.42×)
- Discovered $G = \langle A_0, A_1 \rangle = GL_2(\mathbb{F}_p)$ for $p \in \{5,7,11,13\}$ (full surjectivity) with 100% irreducibility verification
- Validated threshold on real ciphers: Trivium, ChaCha20, and LFSRs (6/6 correct predictions, 100% accuracy)
- Implemented in Python (NumPy, PyTorch, Numba JIT) with complete 140-page LaTeX paper and reproducible code
- **GitHub:** github.com/CoderAwesomeAbhi/neural-cryptanalysis

COMPETITIONS & AWARDS

Mathematics

- **AIME Qualifier** | American Invitational Mathematics Examination | 2026
- **AMC 10 Distinction** | Mathematical Association of America | 2026
- **MATHCOUNTS State Competitor** | Texas State Competition | 2026
- **Stanford Math Tournament (SMT) General:** Top 10% | 2026

Computer Science

- **USACO Silver Division** | USA Computing Olympiad | 2025–2026

Physics

- **USAPHO Qualifier** | USA Physics Olympiad | 2026

Science & Engineering

- **GARSEF 3rd Place Embedded Systems** | Engineering & General Sciences Division | 2026
- **GARSEF 4th Place** | Energy: Sustainable Materials and Design | 2024

TECHNICAL SKILLS

Programming: Python (advanced), Rust (intermediate), C/C++ (intermediate), JavaScript

ML/AI: PyTorch, TensorFlow, scikit-learn, Keras

Scientific Computing: NumPy, SciPy, Matplotlib, Numba (JIT compilation)

Mathematical Tools: LaTeX, Jupyter, Mathematica

Development: Git, GitHub, Linux, VS Code, Docker

RELEVANT COURSEWORK & SELF-STUDY

Completed: Geometry (two grades ahead in math), Computer Science (Python, C++, algorithms)

Self-Study: p-adic Number Theory (Gouvêa's *p-adic Numbers: An Introduction*), Abstract Algebra, Number Theory (Hensel's Lemma, arithmetic dynamics, monodromy theory), Cryptography (pseudorandom generators, linear complexity, Berlekamp-Massey algorithm), Measure Theory (ergodicity, p-adic measures)

RESEARCH STATEMENT

I am passionate about applied mathematics, particularly number theory and algebraic structures, as well as systems security and kernel-level programming. I'm eager to learn from experienced researchers and contribute to ongoing research projects.

What I Seek: Research mentorship where I can learn rigorous mathematical techniques and contribute to your research program. I'm open to working on any project you suggest—I want to learn how professional mathematical research is conducted.

What I Offer: Strong work ethic, solid programming skills (Python, Rust, C), and genuine enthusiasm for mathematics and computer science. I'll do all coding, computations, and writing you need. I'm here to learn and contribute to your research goals.

Last Updated: May 13, 2026